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FedARA: Resource-adaptive Low-rank Personalized Federated Learning via Anchor-driven Representation Alignment on Heterogeneous Edge Devices

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Abstract

*Personalized Federated Learning (PFL) has gained significant attention for enabling participating clients to train customized personalized models on non-IID local data. However, current PFL methods mainly suffer from two limitations: 1) Only the personalized part supports heterogeneous design, while the shared part must remain homogeneous. 2) The semantic representations of models generated on different clients with non-IID data characteristics inevitably tend to be inconsistent, negatively impacting model performance. To conquer them, this paper proposes a novel resource-adaptive personalized **Fed**erated Learning via **A**nchor-driven **R**epresentation **A**lignment (**FedARA**). Concretely, we design a low-rank decomposition and reconstruction fusion scheme for shared feature extractors based on the matrix decomposition technology, where each client can autonomously set the rank value based on its locally available resources, controlling the complexity of extractors and naturally reducing communication and computational costs. Moreover, to address the inconsistency of feature spaces across clients, an anchor-driven representation consistency learning mechanism is developed, which can guide client models to learn unified feature representations and alleviate global knowledge forgetting, thereby improving personalized model performance. Extensive experimental results demonstrate that our method significantly outperforms seventeen state-of-the-art baselines in diverse heterogeneous scenarios with less communication and computational costs.*

1. Introduction

Federated Learning (FL) [15, 22, 35] provides a privacy-preserving distributed learning paradigm to enable cross-edge collaborative intelligence. However, in real-world IoT systems, the efficacy of FL is usually challenged by the heterogeneity in both data distribution and device resources at different IoT clients. Firstly, edge devices owned by clients typically operate in diverse environments, where local data across clients exhibits unbalanced and non-independent identically distributed (non-IID) characteristics [17]. Secondly, clients vary in computational capacity, memory and bandwidth. Thus, conventional FL methods such as FedAvg [22] that require all clients to train an identical homogeneous model structure are often constrained by the capabilities of the least-capable clients, compromising model performance while simultaneously wasting the resources on well-equipped clients. Particularly, it is difficult to guarantee that the trained global model can perform well on each client.

To address heterogeneous challenges, Personalized Federated Learning (PFL) [4, 21, 28, 43] has emerged, aiming to enable each client to train a unique local model tailored to its specific data distribution or even device resources. A prevalent trend is parameter decoupling [1, 18], which separates model parameters into personalized and shared components. The personalized parameters remain locally with private data, while the shared parameters are aggregated to facilitate cross-client knowledge fusion. This paradigm balances global generalization with local personalization, which also reduces communication overhead by transmitting only a subset of parameters.

Nevertheless, in model-decoupled PFL [3, 23, 34], the shared components (typically feature extractors) are prone to drift under non-IID data, causing inconsistent representations in semantic spaces [6, 44], and biased decision boundaries in personalized components (typically classifiers). Aggregating these skewed extractors on server further degrades global generalization. In turn, the resulting aggregated parameters, when used to update local shared components, can potentially create a vicious cycle. Moreover, the full knowledge of global feature extractor is solely used to initialize the local feature extractor, and it tends to degrade during subsequent training batches, leading to catastrophic forgetting of global knowledge. Thus, such inconsistent and incomplete global knowledge transfer ultimately has a negative impact on personalized model performance at clients.

Additionally, most model-decoupled architectures en-

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force homogeneity in the shared component to enable cross-client knowledge fusion, thereby inheriting the fundamental limitations of overall model-homogeneous FL mentioned above. Recent methods such as pFedAFM [39] and pFedES [38] attempt to overcome this by introducing an additional homogeneous small feature extractor in each client and sharing it for cross-client interaction. However, they require training the small shared extractor and local personalized model sequentially at each client to facilitate bidirectional knowledge transfer. This significantly increases computational overhead and training costs for resource-constrained client devices, and the knowledge transfer capability of the shared small homogeneous feature extractor is limited.

To address the limitations of existing works, this paper proposes *resource-adaptive Federated Learning via Anchor-driven Representation Alignment (FedARA)*, a novel PFL framework designed to tackle dual-heterogeneous challenges while alleviating computational and communication costs. Firstly, to accommodate diverse complexity requirements across resource-heterogeneous clients, we introduce a low-rank decomposition and reconstruction fusion scheme based on matrix decomposition technology, where clients can autonomously set the rank value according to locally available resources, thereby controlling model complexity. Compared to the full model, its low-rank versions are more lightweight, reducing the communication and computational costs. Moreover, to mitigate feature space inconsistency across clients, we design an anchor-driven representation consistency learning mechanism to facilitate consistent global knowledge transfer and enhance personalized model performance. Our main contributions are summarized as follows:

- We propose a novel dual-heterogeneous PFL named FedARA for achieving both data-level and model-level personalization in edge intelligence scenarios by leveraging the model-decoupled low-rank architecture.
- A low-rank decomposition and reconstruction fusion scheme for heterogeneous clients is designed, which enables heterogeneous clients to train appropriate local models with diverse complexities based on their practical device resources and achieves the knowledge aggregation across heterogeneous clients.
- An anchor-driven representation consistency learning mechanism is developed, which constructs cross-client consistency anchors and introduces a feature alignment constraint. This effectively mitigates feature distribution drift caused by non-IID data characteristics and alleviates generalization knowledge forgetting, ensuring the effectiveness of cross-client knowledge transfer.
- Extensive experimental results demonstrate that our method significantly improves personalized classification performance across seventeen state-of-the-art (SOTA) baselines and exhibits stronger superiority in addressing dual heterogeneity challenges with low communication and computational costs.

2. Related Work

PFL offers a fresh perspective on addressing the heterogeneous challenges of data distribution or even device resources by focusing on the personalization of participating clients, which typically includes the following categories:

PFL with Submodels. These methods such as HeteroFL [5], FedDropout [32], SRP-pFed [8], and FedSPU [24] generate submodels with varying sizes by pruning or freezing partial neural parameters of initial model to accommodate diverse training requirements of heterogeneous clients. However, due to fragmented and uneven updating of parameters, convergence becomes unstable, particularly under non-IID data distributions.

PFL with Knowledge Distillation. Most of these methods (Fed-ET [2], FedMD [14], FedDF [19] and ACE-pFL [7]) achieve knowledge fusion and integration among heterogeneous models via knowledge distillation using a public dataset. However, obtaining a suitable public dataset is not always feasible. Although subsequent research studies (FedKTL [41], FedGen [45]) attempt to introduce zero-shot knowledge distillation techniques to generate synthetic samples to assist knowledge transfer, their effectiveness remains highly dependent on the quality of the generated samples.

PFL with Mutual Learning. Some studies (FML [27], FedMRL [37], FedKD [33], and pFedAFM [39]) employ a dual model architecture based on mutual learning to address the heterogeneous challenge of device resources, where each client is configured with a shared homogeneous model and a personalized heterogeneous model. The cross-client knowledge transfer is achieved through interaction via homogeneous models. However, each resource-constrained client requires sequentially training two models, which imposes an additional computational burden and is often unacceptable.

PFL with Logits or Prototypes. Some methods (FD [9], FedProto [29], FedSA[44] and FedTGP [42]) upload the logits, representations or prototypes of the local seen-class samples to the server for aggregation, and then return them to guide local training. Uploading this sensitive information may pose a risk of privacy disclosure.

PFL with Model Decoupling. Some methods (LG-FedAvg [18], FedGH [36]) enable heterogeneous architectures by decoupling local models into heterogeneous feature extractors and homogeneous classifiers, thus enabling knowledge exchange across clients through sharing these classifiers. However, constrained by insufficient shared parameter capacity, the efficiency of cross-client knowledge transfer is limited. Conversely, some others such as FedPer [1], FedRep [3], FedBABU [25], and FedAS [34] split the model parameter into a shared feature extractor and a personalized classifier, while the inconsistent and incomplete global knowledge transfer introduced before will harm the personal-

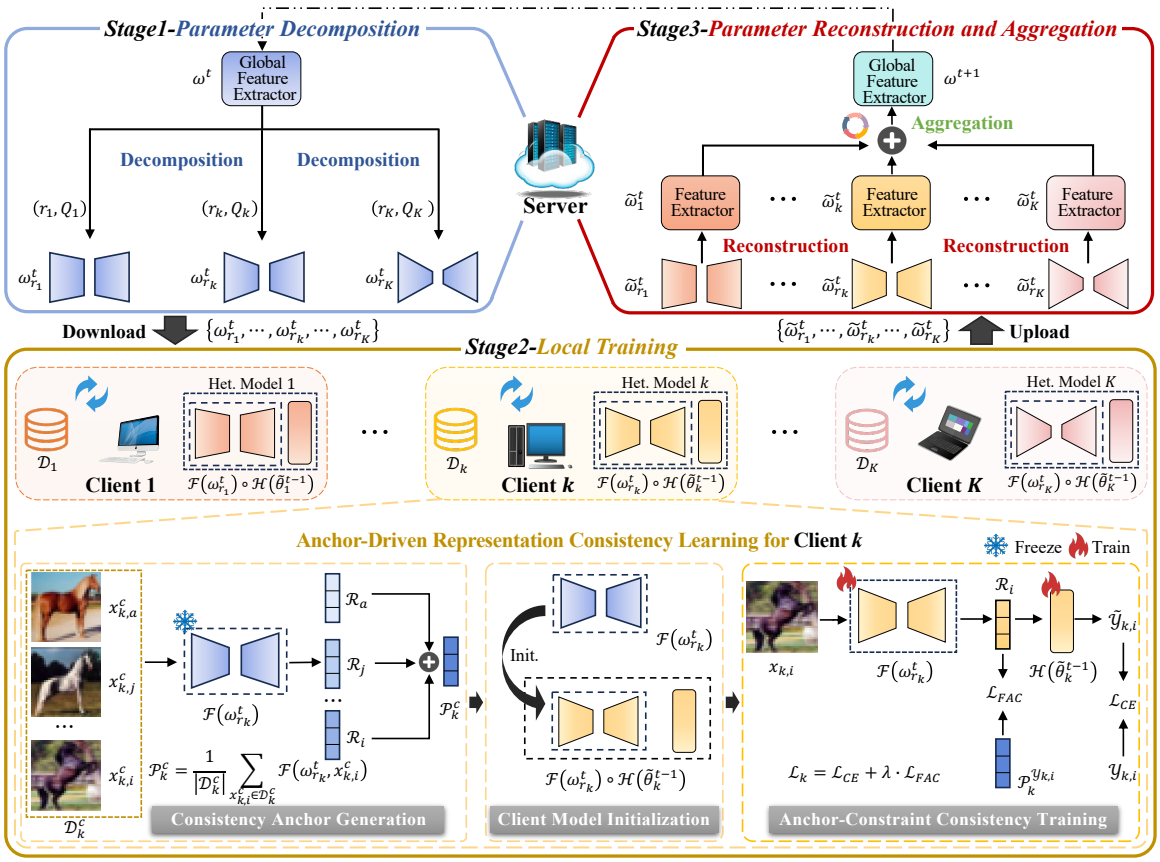


Figure 1. Framework of FedARA

ized model performance. However, all these methods require the shared part to remain homogeneous. This limits their applicability in heterogeneous edge scenarios.

Our Insight. Compared with existing works, FedARA comprehensively meets data-level and model-level personalization requirements, while resolving inconsistent and incomplete global knowledge transfer. Notably, it innovatively leverages low-rank decomposition and reconstruction properties to overcome the homogeneous constraint of the shared part in model-decoupled PFL, while most works [20, 26, 33] predominantly use decomposition for model compression to reduce FL resource consumption. FedARA retains this advantage while offloading the decomposition and reconstruction operations to server without imposing extra computational burdens on clients.

3. Preliminary

3.1. Low-Rank Decomposition of Neural Networks

Fully Connected Layers. Given a fully connected (FC) layer $z' = \sigma(Wz)$, an n-dimensional input z can be mapped to an m-dimensional output z'. Here, σ is a nonlinear activation function, $W \in \mathbb{R}^{m\times n}$ represents the parameters of this layer. To reduce both the multiplication complexity and memory complexity from $O(mn)$ to $O(mr+nr)$, W can be decomposed into the product form UV^T, where $U \in \mathbb{R}^{m\times r}$ and $V \in \mathbb{R}^{n\times r}$, and rank $r \ll \min\{m,n\}$. Then, the computation for the FC layer can be expanded into two low-rank matrix operations: $z' = \sigma(UV^T z)$.

Convolutional Layers. Let $W \in \mathbb{R}^{c_{\text{out}}\times c_{\text{in}}\times\xi\times\xi}$ denote the weights of a convolutional layer, where C_{in} and C_{out} are the input and output channels, and $\xi\times\xi$ is the kernel size. Following [10], W is reshaped into $W' \in \mathbb{R}^{c_{\text{out}}\xi\times c_{\text{in}}\xi}$ and factorized as $W' = UV^T$, where $U \in \mathbb{R}^{c_{\text{out}}\xi\times r}$ and $V \in \mathbb{R}^{c_{\text{in}}\xi\times r}$. U and V are then reshaped into 4D convolutional weights $V \in \mathbb{R}^{r\times c_{\text{in}}\times 1\times\xi}$ and $U \in \mathbb{R}^{c_{\text{out}}\times r\times\xi\times 1}$, replacing the original convolution. This reduces the storage complexity from $O(\xi^2 c_{\text{in}}c_{\text{out}})$ to $O(\xi r(c_{\text{in}}+c_{\text{out}}))$, with $r \le \min\{c_{\text{out}}\xi, c_{\text{in}}\xi\}$. FedARA is also compatible with other low-rank decomposition methods [13, 30, 31].

Regularisation of Decomposed Layers. To ensure the performance of the low-rank decomposition model during training, we incorporate a Frobenius decay regularization, as proposed in [10], by applying a Frobenius norm penalty

$\|U_q V_q^T\|_F^2$ to each factorized layer q.

4. Methodology

4.1. Problem Formulation

In FedARA, we consider a model-decoupled PFL system with a central server and K clients. Each client k ($k = 1, 2, \ldots, K$) holds a private dataset $\mathcal{D}_k$ and a local model $F(\mathcal{W}_k)$. The local model can be decomposed into two parts: $F(\mathcal{W}_k) = \mathcal{F}(\omega_k) \circ \mathcal{H}(\theta_k)$, where $\circ$ denotes the concatenation of model components. Here, $\mathcal{F}(\omega_k)$ is a feature extractor (FE) used for cross-client knowledge interaction, mapping input sample $x \in \mathbb{R}^I$ to corresponding feature representation $\mathcal{R} \in \mathbb{R}^F$. $\mathcal{H}(\theta_k)$ is a personalized classifier adapted to client k's local data distribution, mapping $\mathcal{R}$ to the final prediction $\widetilde{y} \in \mathbb{R}^C$.

4.1.1. Resource-Adaptive Model Architecture Selection

To accommodate the varying demands for model complexity across resource-heterogeneous clients, each client constructs a low-rank heterogeneous model of adaptive complexity based on the global feature extractor (GFE) $\mathcal{F}(\omega)$ prior to federated training. Specifically, let the architecture of the GFE be defined as $\mathcal{F}(\omega) = \{W_1, W_2, \ldots, W_L\}$. Each client k selects, in an independently configurable manner, a subset of layers $Q_k \subseteq \{1, 2, \ldots, L\}$ with larger parameter scales to decompose, which usually have more redundant parameters. Based on its available system resources, client k determines a uniform scaling factor r_k and decomposes each selected layer W_q ($q \in Q_k$) into low-rank matrices U_q and V_q. Through this process, the client constructs a resource-adaptive FE $\mathcal{F}(\omega_{r_k})$, which is then combined with its personalized classifier $\mathcal{H}(\theta_k)$ to form the final heterogeneous low-rank model: $F(\mathcal{W}_k) = \mathcal{F}(\omega_{r_k}) \circ \mathcal{H}(\theta_k)$. Subsequently, each client uploads its structural parameters (r_k, Q_k) to the server, which serves as the key foundation for knowledge fusion among heterogeneous low-rank models.

4.2. Solution Overview

In subsequent rounds of federated training (i.e., communication round $t > 1$), the detailed procedure of FedARA is illustrated in Fig. 1 and described as follows:

1. The server randomly selects a set of clients S^t from the K clients to participate in training at round t. For each client $k \in S^t$, the server decomposes the current GFE parameter ω^t into the corresponding low-rank form $\omega_{r_k}^t$ according to the client's structural parameters (r_k, Q_k), and transmits $\omega_{r_k}^t$ to the client k.
2. Upon receiving the low-rank parameters $\omega_{r_k}^t$, client k updates the parameters $(\omega_{r_k}^t, \widetilde{\theta}_k^{t-1})$ with its local dataset based on the anchor-driven representation consistency learning mechanism (see Sec. 4.3), yielding updated parameters $(\widetilde{\omega}_{r_k}^t, \widetilde{\theta}_k^t)$. Subsequently, $\widetilde{\omega}_{r_k}^t$ is uploaded to the server for cross-client knowledge fusion, while $\widetilde{\theta}_k^t$ is retained locally to preserve the characteristics of the local data distribution.
3. After receiving $\{\widetilde{\omega}_{r_k}^t\}_{k \in S^t}$ uploaded by clients, the server can use the reconstruction operation $R(\cdot)$ to recover each low-rank parameters $\widetilde{\omega}_{r_k}^t$ into a full-rank status $\widetilde{\omega}_k^t = R(\widetilde{\omega}_{r_k}^t)$, whose dimensions align with those of the GFE parameter. The reconstructed parameters $\{\widetilde{\omega}_k^t\}_{k \in S^t}$ are then aggregated to achieve cross-client knowledge fusion, producing the updated GFE parameters ω^{t+1} for the next round. The reconstruction and cross-client knowledge fusion scheme are detailed in Sec. 4.4.

This process is iterated until all client models converge. The complete workflow of FedARA is shown in Algorithm 1. Note that, when $t = 1$, since no globally aggregated FE parameters have been generated yet, the local training on each client proceeds without anchor constraints.

4.3. Anchor-Driven Representation Consistency Learning Mechanism

The core design principle is that the server-aggregated GFE parameters ω embed richer generalization knowledge, enabling more precise feature representations across categories. Accordingly, each client uses the received GFE parameters to construct a *consistency anchor*, which offers a stable reference that guides the local model training and aligns feature representation spaces across clients.

Consistency Anchor Generation. In round t, client k utilizes the low-rank GFE parameters $\omega_{r_k}^t$ to compute consistency anchors for each category c in its local dataset $\mathcal{D}_k$:

$$\mathcal{P}_k^c = \frac{1}{|\mathcal{D}_k^c|} \sum_{x_{k,i}^c \in \mathcal{D}_k^c} \mathcal{F}(\omega_{r_k}^t, x_{k,i}^c), \tag{1}$$

where $\mathcal{P}_k^c$ is the consistency anchor for category c, $\mathcal{D}_k^c \subseteq \mathcal{D}_k$ contains samples of category c, and $|\mathcal{D}_k^c|$ is its size. $\mathcal{F}(\omega_{r_k}^t, x_{k,i}^c)$ extracts high-level semantic representation of sample $x_{k,i}^c$ using frozen parameters $\omega_{r_k}^t$. Since the anchors for category c across different clients are derived from ω^t or its corresponding low-rank approximation $\omega_{r_k}^t$, the set of anchors $\{\mathcal{P}_k^c\}_{k \in S^t}$ for the same category c exhibits intrinsic consistency, thereby providing a unified and stable reference for category c in the feature space.

Client Model Initialization. Client k initializes its local model for training with parameters $\mathcal{W}_k^t = (\omega_{r_k}^t, \widetilde{\theta}_k^{t-1})$, formed by integrating the received parameters $\omega_{r_k}^t$ with its local personalized classifier parameters $\widetilde{\theta}_k^{t-1}$ updated in the previous round.

Anchor-Constraint Consistency Training. In local training, client k processes each sample $(x_{k,i}, y_{k,i}) \in D_k$ through $\mathcal{F}(\omega_{r_k}^t)$ to extract its feature representation $\mathcal{R}_i = \mathcal{F}(\omega_{r_k}^t, x_{k,i})$. To enforce representation consistency, we introduce a Feature Alignment Consistency loss that penalizes

the discrepancy between a sample's feature representation and the consistency anchor of its ground-truth class:

$$\mathcal{L}_{\text{FAC}} = \|\mathcal{R}_i - \mathcal{P}_k^{y_{k,i}}\|_2^2, \tag{2}$$

where $\mathcal{P}_k^{y_{k,i}}$ is the consistency anchor corresponding to the label $y_{k,i}$. The representation $\mathcal{R}_i$ is fed into the personalized classifier to obtain prediction: $\widetilde{y}_{k,i} = \mathcal{H}(\widetilde{\theta}_k^{t-1}, \mathcal{R}_i)$. Therefore, the cross-entropy loss between the predicted and true label is calculated as:

$$\mathcal{L}_{\text{CE}} = \ell(\widetilde{y}_{k,i}, y_{k,i}). \tag{3}$$

Consequently, the total loss function for local model at client k can be formulated as:

$$\mathcal{L}_k = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{FAC}}, \tag{4}$$

where λ is a hyperparameter balancing the two loss terms. The $\mathcal{L}_{\text{FAC}}$ serves a dual purpose: aligning feature spaces across clients and mitigating global generalization knowledge forgetting during local training. The consistency anchors inherently embed generalization knowledge from the GFE parameters, facilitating continuous global knowledge transfer to local models. Thus, this operation effectively alleviates catastrophic forgetting of global knowledge when optimizing local model performance.

The client k updates its model via gradient descent:

$$\widetilde{\mathcal{W}}_k^t \leftarrow \mathcal{W}_k^t - \eta \nabla \mathcal{L}_k, \tag{5}$$

where η is the learning rate and $\widetilde{\mathcal{W}}_k^t = (\widetilde{\omega}_{r_k}^t, \widetilde{\theta}_k^t)$ represents the model parameters after local update.

4.4. Low-Rank Parameter Reconstruction and Cross-Client Knowledge Fusion

Low-Rank Parameter Reconstruction. Upon receiving low-rank FE parameters $\{\widetilde{\omega}_{r_k}^t\}_{k \in S^t}$ from clients, the server reconstructs them into full-rank parameters consistent with the GFE structure via the reconstruction function $R(\cdot)$: $\widetilde{\omega}_k^t = R(\widetilde{\omega}_{r_k}^t)$, where $R(\cdot)$ is the inverse of low-rank decomposition. The reconstruction proceeds as follows:

- *For a convolutional layer*, the low-rank parameters $U_q \in \mathbb{R}^{c_{\text{out}} \times r \times \xi \times 1}$ and $V_q \in \mathbb{R}^{r \times c_{\text{in}} \times 1 \times \xi}$ are first reshaped into matrices $U_q \in \mathbb{R}^{c_{\text{out}}\xi \times r}$ and $V_q \in \mathbb{R}^{c_{\text{in}}\xi \times r}$. The full-rank parameter $W \in \mathbb{R}^{c_{\text{out}} \times c_{\text{in}} \times \xi \times \xi}$ is then obtained by computing $U_q V_q^\top$ and reshaping the resulting matrix.
- *For a FC layer*, the reconstruction is more direct: given $U_q \in \mathbb{R}^{c_{\text{out}} \times r}$ and $V_q \in \mathbb{R}^{c_{\text{in}} \times r}$, the full-rank weight matrix $W \in \mathbb{R}^{c_{\text{out}} \times c_{\text{in}}}$ is computed simply as $U_q V_q^\top$.

Cross-Client Knowledge Fusion. The server aggregates the restructured parameters $\{\widetilde{\omega}_k^t\}_{k \in S^t}$ to achieve cross-client knowledge fusion:

$$\omega^{t+1} = \sum_{k \in S^t} \frac{n_k}{n} \cdot \widetilde{\omega}_k^t, \tag{6}$$

where n_k is the number of samples at client k and n is the total samples across selected clients in round t.

Algorithm 1 FedARA

Require: K: number of clients, S^t: set of clients selected in round t, T: number of communication rounds, $\{\mathcal{D}_k\}_{k=1}^K$: local datasets, $\{r_k, Q_k\}_{k=1}^K$: architectural parameters of the low-rank FE on clients, E_l: number of local epochs, λ: constraint strength, η: learning rate.

ServerExecution:

for $t = 1$ **to** T **do**
  $S^t \leftarrow$ Randomly select a subset from K clients
  for each client $k \in S^t$ **in parallel do**
    $\omega_{r_k}^t \leftarrow \text{Decompose}(\omega^t, r_k, Q_k)$
    $\widetilde{\omega}_{r_k}^t \leftarrow$ **ClientUpdate**$(t, k, \omega_{r_k}^t)$
    $\widetilde{\omega}_k^t \leftarrow R(\widetilde{\omega}_{r_k}^t)$
  end for
  $\omega^{t+1} \leftarrow \sum_{k \in S^t} \frac{n_k}{n} \cdot \widetilde{\omega}_k^t$
end for

ClientUpdate$(t, k, \omega_{r_k}^t)$:

for each category $c \in \mathcal{D}_k$ **do**
  $\mathcal{P}_k^c \leftarrow \frac{1}{|\mathcal{D}_k^c|} \sum_{(x_{k,i}^c, y_{k,i}^c) \in \mathcal{D}_k^c} \mathcal{F}(\omega_{r_k}^t, x_{k,i}^c)$
end for
Parameter initialization: $\mathcal{W}_k^t \leftarrow (\omega_{r_k}^t, \widetilde{\theta}_k^{t-1})$
for $e = 1$ **to** E_l **do**
  if $t > 1$ **then**
    $\mathcal{L}_k = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{FAC}}$
  else
    $\mathcal{L}_k = \mathcal{L}_{\text{CE}}$
  end if
  Perform local training using $\mathcal{D}_k$ and $\mathcal{L}_k$: $\mathcal{W}_k^t \leftarrow \mathcal{W}_k^t - \eta \nabla \mathcal{L}_k$
end for
return $\widetilde{\omega}_{r_k}^t$ belonging to the updated $\widetilde{\mathcal{W}}_k^t = (\widetilde{\omega}_{r_k}^t, \widetilde{\theta}_k^t)$

5. Experiments

5.1. Experimental Setup

Datasets. We evaluate FedARA on three benchmark datasets, CIFAR10 [11], CIFAR100 [11] and Tiny-ImageNet [12], which are widely used in SOTA methods. To simulate non-IID data partitions, we utilize two strategies. One is Dirichlet distribution ($Practical$), whose heterogeneity level is controlled by a concentration parameter α, where smaller α implies greater heterogeneity. The other is Pathological distribution ($Pathological$), where each client is assigned 2 classes for CIFAR10 ($Pathological$ (2/10)), 10 classes for CIFAR100 ($Pathological$ (10/100)) or 20 classes for Tiny-ImageNet ($Pathological$ (20/200)). More dataset and data partition details are provided in Appendix Sec. 7.1.

Models. Under the model-homogeneous setting, we adopt a unified 4-layer CNN for all clients as in [34]. We further introduce FedARA*, which adopts a low-rank variant of this CNN model with $r_k = 0.5$. For model-heterogeneous sce-

Table 1. Test accuracy on CIFAR10 under heterogeneous data with homogeneous models. Avg denotes average accuracy.

Methods	*Practical* ($\alpha = 0.1$)			*Practical* ($\alpha = 0.5$)			*Pathological* (2/10)			Avg
	$P = 0.2$	$P = 0.4$	$P = 1.0$	$P = 0.2$	$P = 0.4$	$P = 1.0$	$P = 0.2$	$P = 0.4$	$P = 1.0$	
FedPer [1]	90.02	89.91	89.90	74.08	72.89	72.82	90.61	90.41	90.34	84.55
FedRep [3]	89.59	89.82	89.68	73.10	72.31	72.54	90.67	90.33	90.23	84.25
FedGen [45]	89.07	89.17	89.18	68.88	68.90	68.85	88.47	88.66	88.69	82.20
Ditto [16]	88.90	89.05	89.07	68.56	68.50	68.42	87.90	88.13	88.24	81.86
LG-FedAvg [18]	89.01	89.11	88.82	68.92	68.96	68.83	88.48	88.85	88.88	82.20
FedGH [36]	89.29	89.32	89.30	67.77	68.17	68.51	89.13	89.23	89.32	82.22
FedBABU [25]	89.97	89.61	89.75	76.37	75.84	75.58	90.23	89.94	89.88	85.24
FedProto [29]	84.87	89.83	90.32	67.49	70.40	71.54	88.68	89.13	89.50	82.41
FD [9]	88.81	89.11	89.35	69.15	69.64	69.78	88.89	89.14	89.20	82.56
FedKD [33]	90.16	90.44	90.44	71.74	72.91	73.37	89.96	90.22	90.38	84.40
FML [27]	88.82	89.61	89.65	68.50	69.43	70.68	87.90	88.54	88.84	82.44
FedALA [40]	90.34	90.41	90.22	77.01	76.74	76.72	90.86	90.53	90.60	85.93
FedMRL [37]	89.17	89.13	89.05	68.23	68.45	68.60	89.04	89.22	89.28	82.24
FedTGP [42]	89.29	90.12	90.56	71.35	72.41	72.80	89.17	90.17	90.61	84.05
FedAS [34]	91.42	90.94	90.83	78.71	77.97	77.35	91.43	91.45	90.87	86.77
pFedAFM [39]	89.18	89.37	89.25	68.42	68.41	68.50	88.97	89.04	89.10	82.24
FedARA (Ours)	**91.71** ↑0.29	**92.06** ↑1.12	**91.95** ↑1.12	**80.34** ↑1.63	**80.49** ↑2.52	**80.25** ↑2.90	**92.19** ↑0.76	**92.58** ↑1.13	**92.33** ↑1.46	**88.21** ↑1.44
FedARA* (Ours)	**91.82** ↑0.40	**92.01** ↑1.07	**91.76** ↑0.93	**79.89** ↑1.18	**79.62** ↑1.65	**79.66** ↑2.31	**92.33** ↑0.90	**92.31** ↑0.86	**92.05** ↑1.18	**87.93** ↑1.16

narios, all baselines follow [36] and use five CNNs (CNN1-CNN5) with decreasing complexity from CNN1 to CNN5. In FedARA, we apply low-rank decomposition to CNN1 with rank ratios tailored to dataset: $\{1.0, 0.5, 0.35, 0.25, 0.15\}$ for CIFAR10/100 and $\{1.0, 0.4, 0.35, 0.29, 0.18\}$ for Tiny-ImageNet, to derive five models matching the complexities from CNN1 to CNN5, which are assigned uniformly at random to clients. Full details of model architectures are in Appendix Sec. 7.2.

Baselines. We compare FedARA with seventeen SOTA methods, including six model-homogeneous PFL methods: FedPer [1], FedRep [3], Ditto [16], FedBABU [25], FedALA [40] and FedAS [34], and eleven model-heterogeneous ones: FedGH [36], LG-FedAvg [18], FD [9], FedGen [45], FedProto [29], FedTGP [42], FedKD [33], FML [27], FedMRL [37], pFedAFM [39] and FedSPU [24]. These baselines cover a broad range of representative designs in both model homogeneous and heterogeneous PFL settings.

Implementation Details. All experiments use identical hyperparameter settings for fair comparison. Following [34], we set the number of clients $K = 20$, local epochs $E_l = 5$, batch size $BS = 16$, learning rate $\eta = 5 \times 10^{-3}$, and communication rounds $T = 100$. The client participation rate P is set to $\{0.2, 0.4, 1.0\}$ for model-homogeneous and 1.0 for model-heterogeneous experiments. For FedARA, the anchor-constraint consistency strength λ is separately set to 2, 5 and 20 for CIFAR10, CIFAR100 and Tiny-ImageNet under model-heterogeneous settings, and to 10 for both CIFAR10 and CIFAR100 under model-homogeneous settings. The Frobenius norm decay regularization in the decomposed layer is fixed to 1×10^{-3}. To mitigate undertrained GFE in early rounds, λ increases linearly from 0 to its target over the first 10 rounds before stabilizing.

5.2. Experimental Results

5.2.1. Heterogeneous Data with Homogeneous Models

As shown in Tabs. 1 and 2, FedARA achieves the highest accuracy in all scenarios, surpassing the best baseline FedAS by 1.44% on CIFAR10 and 5.13% on CIFAR100 on average. Notably, even with a lower-complexity homogeneous low-rank model (FedARA*), it also maintains superior performance, demonstrating both effectiveness and efficiency. The results for Tiny-ImageNet are in Appendix Sec. 8.1.

5.2.2. Heterogeneous Data with Heterogeneous Models

As demonstrated in Tab. 3, which evaluates performance under model heterogeneity, FedARA establishes a new SOTA, achieving the highest accuracy. It outperforms the best baseline (FedTGP on CIFAR10 and Tiny-ImageNet, FedKD on CIFAR100) by a notable margin, with an improvement of 1.45% on CIFAR10, 5.79% on CIFAR100 and 9.51% on Tiny-ImageNet in the *Practical* setting. Similar substantial gains are observed under the *Pathological* setting. Furthermore, as illustrated in Fig. 2, FedARA not only reaches a higher final accuracy but also exhibits faster convergence compared to all competing methods. These consistent results across diverse and challenging conditions robustly validate the efficacy of our approach in successfully tackling the intertwined challenges of statistical and model heterogeneity. Additional convergence results are in Appendix Sec. 8.4.

5.3. Efficiency Analysis

To evaluate the efficiency of FedARA, we measure the total communication and computational overhead across all clients when FedARA, FedAS, FedKD, and FedTGP achieve a target accuracy, under a participation rate of $P = 1.0$.

Table 2. Test accuracy on CIFAR100 under heterogeneous data with homogeneous models. "-" indicates method non-convergence.

Methods	*Practical* ($\alpha = 0.1$)			*Practical* ($\alpha = 0.5$)			*Pathological* (10/100)			Avg
	$P = 0.2$	$P = 0.4$	$P = 1.0$	$P = 0.2$	$P = 0.4$	$P = 1.0$	$P = 0.2$	$P = 0.4$	$P = 1.0$	
FedPer [1]	47.56	47.42	47.18	29.37	28.55	26.92	63.62	63.51	63.48	46.40
FedRep [3]	48.12	47.95	48.01	28.70	28.21	27.42	64.69	64.43	64.91	46.93
FedGen [45]	47.82	47.32	47.06	27.81	27.72	26.68	62.07	62.28	61.67	45.60
Ditto [16]	46.30	46.15	46.10	26.05	25.99	26.04	61.38	61.51	61.39	44.54
LG-FedAvg [18]	46.98	47.14	47.06	27.90	27.51	27.04	61.75	62.14	62.11	45.51
FedGH [36]	47.43	47.22	47.27	27.25	27.61	27.35	62.72	62.77	62.87	45.83
FedBABU [25]	45.20	44.90	45.06	28.44	28.35	27.60	61.88	61.64	62.27	45.03
FedProto [29]	40.19	49.28	51.76	27.51	31.74	33.02	64.52	66.23	66.61	47.87
FD [9]	48.12	48.20	48.26	28.25	28.89	28.51	63.29	64.08	64.29	46.87
FedKD [33]	51.28	52.73	53.43	31.91	33.81	34.66	66.70	67.69	68.53	51.19
FML [27]	45.63	44.50	44.02	24.60	25.85	–	61.67	60.09	60.31	45.83
FedALA [40]	51.67	51.81	51.39	37.18	36.83	36.47	64.69	64.83	65.02	51.09
FedMRL [37]	47.38	47.02	47.14	27.25	27.36	26.76	63.21	62.87	62.81	45.75
FedTGP [42]	49.72	53.50	53.85	31.50	34.05	33.95	67.13	68.49	68.51	51.18
FedAS [34]	57.67	57.06	57.02	39.90	38.39	36.61	70.82	70.46	70.38	55.36
pFedAFM [39]	46.78	46.73	46.84	26.43	26.63	26.43	62.63	64.08	64.29	45.64
FedARA (Ours)	**61.26** ↑3.59	**61.24** ↑4.18	**61.42** ↑4.40	**46.61** ↑6.71	**47.12** ↑8.73	**46.28** ↑9.67	**73.21** ↑2.39	**73.64** ↑3.18	**73.66** ↑3.28	**60.49** ↑5.13
FedARA* (Ours)	**59.84** ↑2.17	**60.67** ↑3.61	**60.29** ↑3.27	**45.28** ↑5.38	**45.44** ↑7.05	**45.18** ↑8.57	**73.13** ↑2.31	**72.90** ↑2.44	**72.39** ↑2.01	**59.45** ↑4.09

Table 3. Test accuracy under model heterogeneity. TiIM denotes Tiny-ImageNet.

Methods	*Practical* ($\alpha = 0.1$)			*Pathological*		
	CIFAR10	CIFAR100	TiIM	CIFAR10	CIFAR100	TiIM
FedGH [36]	87.97	44.25	25.17	88.48	59.69	31.38
LG-FedAVG [18]	88.03	44.37	24.79	88.49	59.66	31.14
FML [27]	88.73	–	–	88.37	–	–
FedGen [45]	87.95	44.04	24.26	88.18	59.44	31.46
FD [9]	88.23	45.76	26.27	88.22	61.67	32.00
FedProto [29]	88.70	48.32	28.22	88.82	63.64	34.37
FedMRL [37]	88.20	44.74	26.11	88.08	60.15	32.70
FedKD [33]	89.05	48.75	–	89.17	63.99	–
FedTGP [42]	89.39	48.70	29.19	88.92	64.57	35.49
FedSPU [24]	88.49	41.97	19.75	86.58	56.10	23.70
pFedAFM [39]	88.35	44.40	25.15	88.27	60.21	32.10
FedARA (Ours)	**90.84**	**54.54**	**38.70**	**91.23**	**70.25**	**41.14**
Δ	↑1.45	↑5.79	↑9.51	↑2.06	↑5.68	↑5.65

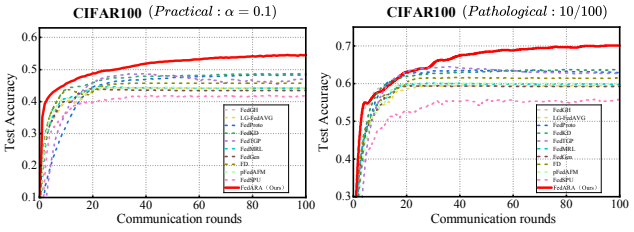


Figure 2. Accuracy vs. communication rounds under model heterogeneity.

Computational Overhead. As illustrated in Fig. 3, under model-homogeneous scenarios, FedARA significantly reduces computational costs compared to the optimal baseline FedAS (see Tabs. 1 and 2). FedARA* further lowers computational overhead by leveraging low-rank operations. In heterogeneous models, FedARA exhibits lower computational overhead than both FedTGP and FedKD (see Tab. 3).

Communication Cost. Fig. 4 shows that in homogeneous settings, FedARA substantially reduces communication cost relative to FedAS, thanks to the anchor-driven representation alignment mechanism that accelerates convergence. FedARA* further cuts communication by transmitting only low-rank FE parameters. In heterogeneous scenarios, FedARA incurs moderately higher communication cost than FedTGP and FedKD, which exchange only local

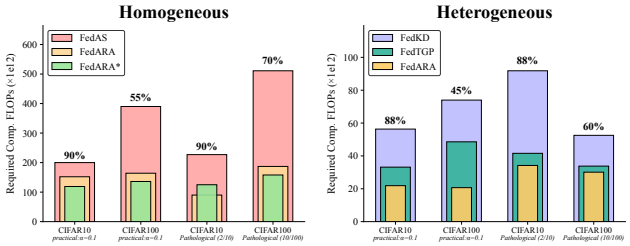


Figure 3. Total computational cost at target accuracy.

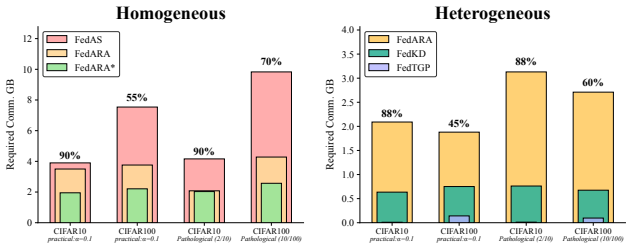


Figure 4. Total communication cost at target accuracy.

seen-class average representations or approximate gradients, but achieves significantly higher accuracy (see Tab. 3). Nevertheless, FedARA remains more communication-efficient than other PFL methods that transmit full model parameters (e.g., FedALA), while achieving superior accuracy.

Comprehensive comparisons with other baselines are provided in Appendix Sec. 8.6 (see Tabs. 9 to 12).

5.4. Ablation Study on the Constraint Strength

We conduct an ablation study on the strength of anchor constraint λ ($\lambda \in \{0, 0.1, 0.5, 1, 2, 5, 10, 20\}$) under heterogeneous models to analyze the effectiveness of the anchor-driven representation consistency learning mechanism. As shown in Fig. 5, removing the constraint ($\lambda = 0$) causes a significant accuracy drop in FedARA, indicating that local models are severely affected by non-IID data. This leads to substantial bias in the learned representation space and degrades overall model performance, thereby confirming the necessity of anchor constraint. As λ increases, both the convergence speed and final accuracy improve, demonstrating that stronger anchor constraint promotes cross-client feature alignment. However, excessively large values (e.g., $\lambda = 20$) over-regularizes local model training, suppressing personalized knowledge extraction and slowing convergence. See Appendix Sec. 8.2 for extended ablation studies.

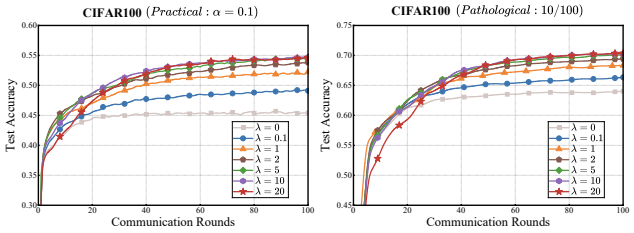


Figure 5. Ablation study on the anchor-constraint strength λ.

5.5. Visualization for Anchor Constraint

To better illustrate the role played by anchor constraint, we visualize the feature space of each client on CIFAR100 under the model-homogeneous setting using t-SNE, thereby avoiding potential feature dispersion caused by model heterogeneity and low-rank decomposition. As shown in Fig. 6, without the anchor constraint ($\lambda = 0$), features from different clients are highly entangled with blurred class boundaries, indicating significant inconsistency in the feature spaces learned by local models. In contrast, with the anchor constraint, features across clients form compact and well-separated clusters, resulting in a semantically coherent and highly discriminative embedding space. These visual results align with our quantitative analysis. Additional visualizations, including baseline comparisons and fine-grained results, are provided in Appendix Sec. 8.3.

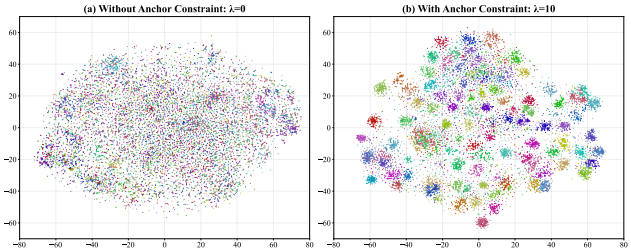


Figure 6. Visualizations with and without anchor constraint.

5.6. Robustness to Data Heterogeneity

As shown in Tab. 4, FedARA achieves the highest accuracy across all baselines, demonstrating strong robustness to non-IID data distribution. Moreover, as the α or classes increases (indicating reduced heterogeneity), the performance of all compared methods, including FedARA, deteriorates with a corresponding decline in personalized learning effectiveness. This finding aligns with the observations reported in [27]. The result for CIFAR10 is in Appendix Sec. 8.5.

Table 4. Robustness to non-IID data on CIFAR100.

Methods	*Practical*			*Pathological*		
	0.1	0.5	1.0	10/100	30/100	50/100
FedGH [36]	44.25	24.98	18.55	59.69	37.57	26.22
LG-FedAVG [18]	44.37	24.74	18.51	59.66	37.59	26.34
FML [27]	–	–	–	–	35.11	–
FedMRL [37]	44.74	25.97	18.43	60.15	38.19	27.09
FedGen [45]	44.04	24.77	18.34	59.44	37.37	26.42
FD [9]	45.76	26.17	19.87	61.67	39.27	27.38
FedProto [29]	48.32	28.35	20.23	63.64	40.43	27.24
FedKD [33]	48.75	28.52	20.38	63.99	42.06	31.23
FedTGP [42]	48.70	28.82	21.61	64.57	40.03	27.10
FedSPU [24]	41.97	25.03	20.14	56.10	38.07	25.36
pFedAFM [39]	44.40	24.34	18.26	60.21	37.47	26.46
FedARA (Ours)	**54.54**	**38.27**	**30.41**	**70.25**	**49.43**	**41.02**
Δ	↑5.79	↑9.45	↑8.80	↑5.68	↑7.37	↑9.79

6. Conclusion

This paper proposes a novel resource-adaptive low-rank PFL framework FedARA, capable of achieving both model-level and data-level personalization. Specifically, a low-rank decomposition and reconstruction fusion scheme is designed to meet the diverse complexity requirements of model training at resource-heterogeneous clients by autonomously setting the rank value. Moreover, we present an anchor-driven representation consistency learning mechanism to guide local models at different clients with non-IID characteristics to learn consistent representation and mitigate the problem of generalization knowledge forgetting. The effectiveness of FedARA has been validated with seventeen SOTA benchmarks over various heterogeneous scenarios.

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